

MARLA, Australia

WMO: 944770

Lat: 27.300S

Lon: 133.620E

Elev: 324

StdP: 97.49

Time Zone: 9.50 (AUA)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.2	3.7	-9.7	1.7	21.3	-7.9	2.0	22.2	9.4	15.0	7.8	19.2	0.2	310	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	41.5	19.8	40.1	19.3	38.5	18.9	23.6	30.7	22.7	31.4	21.6	31.9	3.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.3	26.6	20.3	15.6	26.3	18.5	13.9	26.0	72.5	30.4	68.9	31.7	64.7	31.8	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.5	6.5	-0.7	44.1	1.5	1.0	-1.8	44.8	-2.6	45.4	-3.4	46.0	-4.5	46.7		
(5)			-2.3	25.3	1.9	1.7	-3.6	26.6	-4.7	27.6	-5.8	28.6	-7.1	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	30.0	29.4	25.5	20.2	16.0	12.7	12.4	14.1	19.5	21.8	25.3	27.9	(6)
(7)		DBStd	7.24	3.73	3.92	3.92	3.36	3.74	3.32	2.90	3.37	4.04	4.45	4.32	3.99	(7)
(8)		HDD10.0	18	0	0	0	0	0	8	8	3	0	0	0	0	(8)
(9)		HDD18.3	661	0	0	1	16	94	176	187	138	33	14	2	0	(9)
(10)		CDD10.0	4104	622	543	481	307	187	87	81	130	286	365	459	556	(10)
(11)		CDD18.3	1703	363	309	224	73	23	5	1	7	68	121	212	297	(11)
(12)		CDH23.3	23334	5427	4562	2799	826	231	40	22	117	863	1550	2712	4184	(12)
(13)	CDH26.7	13204	3458	2838	1454	280	56	7	2	24	361	749	1478	2497	(13)	
(14)	Wind	WSAvg	2.3	2.7	2.6	2.2	1.8	1.6	1.4	1.7	2.2	2.6	2.9	2.8	2.7	(14)
(15)	Precipitation	PrecAvg	236	19	36	13	13	16	14	11	9	12	21	22	33	(15)
(16)		PrecMax	400	88	180	63	83	63	81	87	64	53	73	78	127	(16)
(17)		PrecMin	106	3	1	1	0	2	1	0	1	0	0	0	0	(17)
(18)		PrecStd	95	22	46	17	21	17	18	19	14	16	22	21	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	42.9	39.4	34.7	30.9	28.1	26.1	30.2	36.1	39.1	41.5	41.9	(19)
(20)			MCWB	20.5	21.2	19.8	17.0	15.7	15.1	12.9	15.0	16.4	16.9	18.4	18.9	(20)
(21)		2%	DB	41.6	40.9	37.0	31.8	28.3	24.7	23.6	26.8	33.6	36.0	38.5	40.1	(21)
(22)			MCWB	20.3	19.8	19.6	15.9	15.1	14.2	12.3	13.3	15.5	17.1	18.1	18.8	(22)
(23)		5%	DB	40.1	39.5	35.4	30.0	26.2	22.0	21.6	24.4	31.2	33.5	36.4	38.2	(23)
(24)			MCWB	19.7	19.3	18.9	15.1	14.3	12.7	11.6	12.5	15.2	16.0	17.4	18.2	(24)
(25)		10%	DB	38.2	37.8	33.5	28.1	24.1	19.9	19.5	22.0	28.4	30.8	34.2	36.5	(25)
(26)			MCWB	19.0	19.4	17.6	15.1	13.6	11.9	10.6	11.2	14.2	15.2	16.9	17.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	24.8	23.1	20.6	17.2	17.0	16.7	16.2	18.2	20.0	21.6	23.4	(27)
(28)			MCDWB	30.3	30.1	31.1	26.1	27.8	23.5	21.2	26.5	29.9	31.3	30.6	31.1	(28)
(29)		2%	WB	23.6	23.8	21.8	19.3	15.9	15.7	14.3	14.5	17.2	18.7	20.2	22.0	(29)
(30)			MCDWB	29.1	31.6	31.5	24.2	26.0	21.5	19.6	24.1	27.6	29.7	29.7	31.2	(30)
(31)		5%	WB	22.2	23.1	20.7	17.6	15.1	14.1	12.6	13.2	15.9	17.2	19.5	20.8	(31)
(32)			MCDWB	32.8	32.1	30.7	25.7	23.5	21.1	19.8	22.0	28.1	27.9	30.2	30.3	(32)
(33)		10%	WB	20.9	22.1	19.4	16.4	14.4	12.6	11.3	12.0	15.0	16.2	18.7	19.6	(33)
(34)			MCDWB	34.1	31.6	30.2	25.4	22.4	18.6	18.1	19.9	27.2	27.8	29.9	30.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.0	12.5	12.7	13.4	12.9	12.7	13.1	13.9	14.1	13.7	13.2	12.8	(35)
(36)			MCDBR	14.2	14.6	13.8	15.8	14.9	14.9	16.4	16.5	17.4	16.6	16.1	14.9	(36)
(37)			MCWBR	4.7	4.7	4.7	6.5	6.7	7.3	8.5	7.8	6.4	6.4	5.4	4.9	(37)
(38)		5% WB	MCDWB	10.6	9.6	10.6	11.0	12.3	11.0	12.6	14.1	15.2	13.1	13.0	11.8	(38)
(39)			MCWBR	3.6	4.1	4.2	5.0	5.9	6.1	7.8	6.9	6.2	5.8	4.3	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.362	0.357	0.337	0.322	0.297	0.283	0.286	0.300	0.334	0.348	0.360	0.353	(40)	
(41)		taud	2.478	2.519	2.542	2.562	2.595	2.617	2.592	2.521	2.434	2.422	2.442	2.495	(41)	
(42)		Ebn at Noon	980	969	958	927	909	904	914	937	945	966	975	991	(42)	
(43)		Edh at Noon	118	110	102	91	80	74	79	93	112	120	121	116	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.96	7.37	6.48	5.18	4.15	3.61	3.92	4.98	6.15	7.05	7.56	7.79	(44)	
(45)		RadStd	0.38	0.59	0.40	0.33	0.32	0.25	0.24	0.24	0.30	0.51	0.39	0.44	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page